Molecule identification from MS/MS for CASMI 2026: one change per submission, from 0.115 to 0.356

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Abstract

We rank candidate structures for 400 molecules from their tandem mass spectra in the CASMI 2026 Kaggle competition, scored by MRR@25 on the InChIKey connectivity block. The first submission scored 0.115 against a local estimate of 0.664; the estimate came from labels drawn from the same data as the candidates, and was retracted. Since then every number reported is a public leaderboard score, and each of the 18 submissions changed one thing (a nineteenth re-ran v19 unchanged). A PubChem-scale pool with a popularity prior, a learned spectrum-to-fingerprint model, and two forward models that predict each candidate's spectrum take the score to 0.349; a second PubChem pass that avoids structures similar to the copied train rows takes it to 0.356, against 0.421 for the public notebook these components come from. No local estimate of the score exists.

1 Task

Each test molecule has one or more LC-MS/MS spectra (400 molecules, 1213 spectra). A submission lists up to 25 SMILES per molecule; the score is the mean reciprocal rank of the answer, matched on the first 14 characters of the InChIKey. The competition runs notebooks offline, so every database and model is attached as a dataset. Only the public split of the leaderboard is visible.

2 Method

1. **Candidates.** Within a 5 ppm window of the molecule's neutral mass, the 1000 best-documented PubChem connectivities (by $\log(1+\text{substances})+\log(1+\text{PubMed})$) from a 106-million-structure tier, plus train structures in the window.

2. **Copied spectra.** 1189 of 1213 test spectra are bit-identical to a train row; every molecule has at least one. The train structure behind it is removed from the candidates.

3. **Ranking.** An ensemble of two public FPNet checkpoints predicts fingerprint logits z per spectrum and per polarity-merged spectrum; a candidate with fingerprint f scores $f \cdot z$, z-scored within the molecule, plus $0.25\times$ popularity.

4. **Forward models.** For the top 60, ICEBERG and GLACIER predict every candidate's spectrum. Within each same-formula group reaching the top 25 the order becomes $z(\text{score}) + \lambda\, z(\text{ICEBERG}) + \lambda\, z(\text{GLACIER})$, with $\lambda = 1$.

3 The funnel

stage	surviving
test molecules	400
candidates per molecule (median, max)	1038, 1445
copied train structures removed	400 (one per molecule)
molecules with $[M+H]^+$ / $[M+Na]^+$ spectra (forward-model coverage)	371
candidates scored by each forward model	21,865 of 23,552 sent
molecules whose top-1 the forward models change ($\lambda = 1$)	208

4 Results

Every score is the public leaderboard's. Fourteen submissions were made; all were scored; none is omitted.

v	change from the previous kept version	public MRR@25
1	fragment explainability over train's 275,798 structures	0.115
4	+ PubChem tier, 4000 nearest-mass per molecule	0.049
5	popularity shortlist of 300, +0.25× popularity	0.135
6	FPNet in place of fragment explainability	0.250
7	FPNet ensemble (two checkpoints)	0.258
8	popularity weight 0.15 (not kept)	0.254
9	shortlist 1000	0.261
11	copied train structures removed	0.262
12	ICEBERG, CPU, 269 of 371 molecules scored	0.286
13	ICEBERG on a T4, all 371	0.292
14	+ GLACIER, $\lambda = 0.5$	0.320
15	$\lambda = 1.0$	0.349
16	GLACIER on $[M+H]^+$ only (not kept)	0.345
17	$\lambda = 1.5$ (not kept)	0.346
18	+ spectrum-driven second PubChem pass (not kept)	0.353
19	the same, skipping candidates with Tanimoto ≥ 0.5 to the copied structure	**0.356**
19	identical re-run	0.353
20	threshold 0.4 (not kept)	0.354
21	ICEBERG weight 0.5, GLACIER 1.0 (not kept)	0.333

Two levers account for almost all of the gain: the learned spectrum model (+0.115, v5 to v6) and the forward models (+0.087, v11 to v15). Tuning the shortlist and the prior's weight (v8, v9) moved the score by under 0.01, as did removing the copied structures, although that changed 267 top-1 picks: FPNet, trained on the copied rows, had returned their structures, and what replaced them was as rarely right. Identical runs differ by about ±0.005; v16, v17 and v18 fall inside it and were reverted. v18 is the instructive one: on 280 held-out train molecules, scored by an FPNet that never saw them, a second PubChem pass chosen by the spectrum lifted the answer's presence among the candidates from 126 to 236 and MRR@25 from 0.241 to 0.405. On the test set it moved the score by +0.004. The production FPNet was trained on the train rows the test spectra copy, and the pass's top-1 picks moved toward those copied structures (median Tanimoto 0.26 to 0.42 over the 192 molecules whose top-1 changed). v19 skips second-pass candidates with Tanimoto ≥ 0.5 to the copied structure (894 dropped over 400 molecules) and scored 0.356, +0.007 over v15. Against v18, which differs only by the filter, the gain is

+0.003, inside the noise, so v19 is kept but does not clear G1; its threshold was also chosen after seeing v18. An identical re-run of v19 scored 0.353, and the 0.4 threshold (v20) 0.354: the filter's effect is not distinguishable from noise.

5 What is not established

- **A local estimate of the score.** Train-derived labels rated our first submission at 0.664 against 0.115 on the leaderboard. Public notebooks cannot supply labels: their downloadable outputs score only about 12 smoke-test molecules and write a placeholder (`CCO`) for the rest, since full predictions run only inside Kaggle's scoring rerun; an earlier reading of "three lineages agreeing on 12 of 400" was those placeholders, as was a local 0.030 for the 0.421 notebook. Nor can train supply a test-like held-out set: only 41 isomer pairs in 2.54 million rows share a spectrum with at least 3 peaks, and 7 of them split across the fold-safe FPNet's training as test answers and copied structures do. Each submission is the only measurement.
- **The private score.** Three choices — the shortlist size, λ, and GLACIER's input — were selected on the public split, as was v19's similarity threshold, so 0.356 is optimistic.
- **Candidate recall.** Whether the answer is among the candidates cannot be separated from ranking with one MRR.
- **Why the data is shaped as it is.** The copied train structures are the answer for at most $\sim$46 of 400 molecules.
- **The environment switch alone.** The move to a pinned Python 3.12 image and RDKit 2026.3 (v12) was not measured separately.
- **29 molecules** receive no forward-model evidence.

Sources

Code and history: `https://github.com/charleneleong-ai/ai4science/pull/80`. Every input and checkpoint with its version: `casmi/docs/experiment-card.md`. Competition: `https://www.kaggle.com/competitions/enveda-CASMI26-molecule-id-mass-spectra`.